

Parameter Configuration.

Dropout = 0.1
SEED = 42
layers = 2
optimizer='adam'
loss='mse'
WINDOW=100
Units=32
Epochs=5
Patience=3
1 Year Data
RMSE (Root Mean Squared Error)
MAE (Mean Absolute Error)

Layer (type)	Output Shape	Param #
lstm (LSTM)	(None, 100, 32)	4,352
dropout (Dropout)	(None, 100, 32)	0
lstm_1 (LSTM)	(None, 32)	8,320
dropout_1 (Dropout)	(None, 32)	0
dense (Dense)	(None, 1)	33
Total params: 12,705 (49.63 KB)		
Trainable params: 12,705 (49.63 KB)		
Non-trainable params: 0 (0.00 B)		

Batch_Size(32)

```
Epoch 1/5
1397/1397 ————— 104s 72ms/step - loss:
0.0116 - val_loss: 1.9982e-04
Epoch 2/5
1397/1397 ————— 139s 69ms/step - loss:
0.0012 - val_loss: 1.5432e-04
Epoch 3/5
1397/1397 ————— 97s 69ms/step - loss:
7.6997e-04 - val_loss: 1.3901e-04
Epoch 4/5
1397/1397 ————— 99s 71ms/step - loss:
4.7728e-04 - val_loss: 1.2989e-04
Epoch 5/5
1397/1397 ————— 98s 70ms/step - loss:
3.3230e-04 - val_loss: 1.2299e-04

1397/1397 ————— 28s 20ms/step - loss:
2.8724e-05
MSE Train: 3.9842689147917554e-05
464/464 ————— 9s 19ms/step - loss:
1.0584e-04
MSE Test: 0.00012298976071178913
464/464 ————— 8s 18ms/step - loss:
9.6633e-05
MSE Val: 0.00010950281284749508

Test Set RMSE (Original Scale): 0.000926 EURUSD
Test Set MAE (Original Scale): 0.000645 EURUSD
Validation Set RMSE (Original Scale): 0.000873 EURUSD
Validation Set MAE (Original Scale): 0.000565 EURUSD
Training Set RMSE (Original Scale): 0.000527 EURUSD
Training Set MAE (Original Scale): 0.000361 EURUSD
```

Batch_Size(16)

```
Epoch 1/5
2793/2793 ————— 233s 82ms/step - loss:
0.0075 - val_loss: 1.7301e-04
Epoch 2/5
2793/2793 ————— 257s 80ms/step - loss:
7.7440e-04 - val_loss: 1.7192e-04
Epoch 3/5
2793/2793 ————— 229s 82ms/step - loss:
3.8425e-04 - val_loss: 1.1308e-04
Epoch 4/5
2793/2793 ————— 232s 83ms/step - loss:
3.0097e-04 - val_loss: 1.7294e-04
Epoch 5/5
2793/2793 ————— 257s 81ms/step - loss:
2.9233e-04 - val_loss: 1.2133e-04

1397/1397 ————— 33s 24ms/step - loss:
3.2397e-05
MSE Train: 5.4904434364289045e-05
464/464 ————— 11s 24ms/step - loss:
1.2963e-04
MSE Test: 0.00012132591655245051
464/464 ————— 11s 24ms/step - loss:
9.1299e-05
MSE Val: 9.73146889009513e-05

Test Set RMSE (Original Scale): 0.000919 EURUSD
Test Set MAE (Original Scale): 0.000716 EURUSD
Validation Set RMSE (Original Scale): 0.000823 EURUSD
Validation Set MAE (Original Scale): 0.000567 EURUSD
Training Set RMSE (Original Scale): 0.000618 EURUSD
Training Set MAE (Original Scale): 0.000454 EURUSD
```

Batch_Size(64)

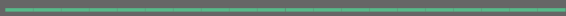
Epoch 1/5

699/699  **77s** 111ms/step - loss: 2.4731e-04 - val_loss: 8.2150e-05

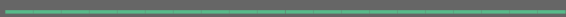
Epoch 2/5

699/699  **78s** 105ms/step - loss: 2.4517e-04 - val_loss: 1.2312e-04

Epoch 3/5

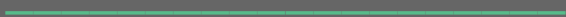
699/699  **72s** 103ms/step - loss: 2.3860e-04 - val_loss: 8.2859e-05

Epoch 4/5

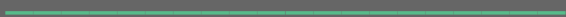
699/699  **87s** 110ms/step - loss: 2.4907e-04 - val_loss: 1.8454e-04

1397/1397  **35s** 25ms/step - loss: 3.0343e-05

MSE Train: 6.422208389267325e-05

464/464  **12s** 25ms/step - loss: 2.5758e-04

MSE Test: 0.00018454364908393472

464/464  **12s** 25ms/step - loss: 1.6622e-04

MSE Val: 0.0001590564352227375

Test Set RMSE (Original Scale): 0.001134 EURUSD

Test Set MAE (Original Scale): 0.000897 EURUSD

Validation Set RMSE (Original Scale): 0.001053 EURUSD

Validation Set MAE (Original Scale): 0.000828 EURUSD

Training Set RMSE (Original Scale): 0.000669 EURUSD

Training Set MAE (Original Scale): 0.000488 EURUSD

Batch_Size(128)

Epoch 1/5

350/350  **50s** 143ms/step - loss: 2.3882e-04 - val_loss: 1.2360e-04

Epoch 2/5

350/350  **80s** 138ms/step - loss: 2.4473e-04 - val_loss: 7.5078e-05

Epoch 3/5

350/350  **83s** 141ms/step - loss: 2.4611e-04 - val_loss: 1.1703e-04

Epoch 4/5

350/350  **53s** 152ms/step - loss: 2.3536e-04 - val_loss: 1.3848e-04

Epoch 5/5

350/350  **80s** 144ms/step - loss: 2.4603e-04 - val_loss: 2.0565e-04

1397/1397  **34s** 24ms/step - loss: 4.2679e-05

MSE Train: 7.887211540946737e-05

464/464  **11s** 25ms/step - loss: 2.9453e-04

MSE Test: 0.00020565417071338743

464/464  **11s** 25ms/step - loss: 2.0324e-04

MSE Val: 0.000191545594134368

Test Set RMSE (Original Scale): 0.001197 EURUSD

Test Set MAE (Original Scale): 0.000957 EURUSD

Validation Set RMSE (Original Scale): 0.001155 EURUSD

Validation Set MAE (Original Scale): 0.000956 EURUSD

Training Set RMSE (Original Scale): 0.000741 EURUSD

Training Set MAE (Original Scale): 0.000570 EURUSD